

Deep Learning in Breast MRI to Determine the Need for Oncotype Testing

Abstract

Objectives

To evaluate whether deep learning applied to breast MRI can determine whether Oncotype testing is needed by predicting binary Oncotype score category, defined as low recurrence score (≤ 18) versus elevated recurrence score (> 18).

Materials and methods

This retrospective study used breast MRI examinations with available Oncotype score labels from the Duke-Breast-Cancer-MRI collection. MRI examinations were processed as three-dimensional NIfTI series input, intensity-normalized, and resized to $256 \times 256 \times 64$. Training was performed using a sensitivity-optimized workflow with five-fold stratified cross-validation, class-balanced weighting, and hyperparameter optimization. Particular attention was given to false-positive samples, as these represent cases in which artificial intelligence would incorrectly suggest the need for Oncotype testing. The number of eligible patients and studies was TBD. External validation was TBD.

Results

The best-performing artificial intelligence configuration was TBD. Cross-validation performance was TBD, including area under the receiver operating characteristic curve, sensitivity, specificity, accuracy, positive predictive value, and negative predictive value. Model calibration and independent test-set performance were TBD.

Conclusion

This study demonstrates that artificial intelligence analysis of breast MRI may support decision-making on whether Oncotype testing is required. The completed testing indicates TBD diagnostic performance, with clinical utility depending on further validation, particularly regarding the rate and consequences of false-positive recommendations for Oncotype testing.

Clinical relevance statement

Artificial intelligence–based assessment of breast MRI could help identify patients unlikely to require Oncotype testing, potentially reducing unnecessary molecular testing while supporting personalized breast cancer risk stratification.

Key Points

- Oncotype testing is clinically important but may not be necessary for all patients with breast cancer.
- Artificial intelligence applied to breast MRI may help identify patients with low versus elevated Oncotype score categories.
- MRI-based artificial intelligence could support personalized decision-making on whether Oncotype testing is required,

Materials and methods

Opis duka

Data loading

Clinical and imaging metadata were obtained from the Duke-Breast-Cancer-MRI collection. Patients without an available Oncotype recurrence score were excluded. Oncotype scores were converted into binary categories, with scores of 18 or lower classified as low and scores above 18 classified as elevated. Patient identifiers were linked to the corresponding MRI study and series identifiers using the collection metadata. MRI series were grouped by examination, and up to five series were retained for each study.

MRI examinations were loaded as three-dimensional NIfTI volumes. Each volume was converted to floating-point format, arranged with an explicit image-channel dimension, and normalized on a per-volume basis. All volumes were resized using trilinear interpolation to a common spatial resolution of $256 \times 256 \times 64$ voxels. Preprocessed volumes were stored in a cache to avoid repeated preprocessing during training.

For examinations containing fewer than five MRI series, zero-valued volumes were added to standardize the number of input series. The resulting volumes were stacked into a five-channel three-dimensional tensor for each examination. Labels were loaded simultaneously and returned as integer class values for binary classification.

GRAF -> jak byl podzielony dataset

Classification

The classifier received the preprocessed three-dimensional breast MRI examinations and generated two output scores corresponding to low Oncotype recurrence score and elevated recurrence score. The class with the highest output score was assigned as the predicted category.

To account for class imbalance, class-specific weights were incorporated into the cross-entropy component of the objective function. Training additionally used a sensitivity-aware penalty designed to reduce false-negative classifications of the elevated-score group. Soft sensitivity was calculated from the predicted probability of the positive class among samples with an elevated Oncotype score. The total training loss was defined as the sum of weighted cross-entropy and the sensitivity penalty multiplied by a tunable weighting coefficient.

TU WZOR? - to pytanie :)

During training and validation, total loss, cross-entropy loss, sensitivity loss, soft sensitivity, and classification accuracy were recorded for each epoch. Validation performance was assessed using the area under the receiver operating characteristic curve, sensitivity, accuracy, confusion matrices, and receiver operating characteristic curves. Model outputs and performance curves were recorded using TensorBoard

Because the intended clinical application was to determine whether Oncotype testing could potentially be avoided, particular attention was given to false-positive predictions. These represented patients with a low Oncotype recurrence score who were incorrectly classified as requiring further Oncotype testing